

HEXADECIMAL & COLOUR CODES

A colourful lesson for young coders



Lesson includes: hex counting • the # symbol • RGB mixing • a guess-the-colour activity • a real-world tour • a recap worksheet

Everything you see on a screen is built out of colour. Screens are covered in thousands of tiny dots called **pixels**, and every single pixel glows with its own tiny mix of light. When we understand how computers describe colour, we can start building our own websites, games, and art — and even teach AI tools what colours to use!

Lesson goals

- Understand what hexadecimal numbers are and why computers like them.
- Read a hex colour code like **#FF5733** and know what each part means.
- Mix Red, Green, and Blue light to predict what colour a code will make.
- Try a hands-on activity guessing colours from their codes.

Quick warm-up question: Can you think of a time you picked a colour on a screen — for a drawing app, a video game character, or a slide background? That colour was almost certainly stored as a hex code, even if you never saw it!

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Counting Beyond 9: What Is Hexadecimal?

Most of the time we count using **10 digits**: 0, 1, 2, 3, 4, 5, 6, 7, 8, 9. This is called the decimal system, because *dec* means ten. Hexadecimal is a different counting system that uses **16 digits** instead — *hexa* means six and *deci* means ten, so $6 + 10 = 16$.

Once a hexadecimal count reaches 9, it keeps going by borrowing letters: A, B, C, D, E, and F. Each letter simply stands in for a two-digit decimal number, as shown below.

Decimal	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Hex	0	1	2	3	4	5	6	7	8	9	A	B	C	D	E	F

A = 10 B = 11 C = 12 D = 13 E = 14 F = 15

Try it yourself

Write the hexadecimal digit for each decimal number. (Answers are on the recap page!)

11 = ____	14 = ____	9 = ____	15 = ____	12 = ____
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Whenever a computer sees the symbol # followed by six letters and numbers, it knows exactly what to do: read this as a colour! The hash symbol is like a little flag that says, “Hey, the code right after me describes a colour, not a word or a regular number.”



A coral-orange colour

#FF5733 is a hex colour code. It always has six characters split into three matching pairs:

- **FF** — the Red amount
- **57** — the Green amount
- **33** — the Blue amount

We'll unpack exactly what those pairs mean on the next page.

Why letters AND numbers? Because hex code pairs run from 00 all the way up to FF (which is 255 in decimal), a single pair needs all 16 hexadecimal digits — 0 through 9 *and* A through F — to count that high in just two characters.

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Three Pairs: Red, Green, Blue

Every six-digit hex colour code splits neatly into three two-digit pairs. Each pair controls one colour of light:

Digits 1–2 Red amount	Digits 3–4 Green amount	Digits 5–6 Blue amount
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The RGB colour model

This is called the **RGB model**, for Red, Green, and Blue — the three primary colours of light. Mixing different amounts of these three lights together can create every colour a screen can show, from sunny yellows to deep purples. Each colour's amount is measured from **00** (none of that light) to **FF** (the maximum amount, equal to 255).

Think of each pair like a volume dial: 00 is silent (no light), and FF is turned all the way up (full brightness).

Worked example: breaking down #FF5733

R		255 (0xFF)
G		87 (0x57)
B		51 (0x33)

Red is at **FF** (full strength), Green is at **57** (medium strength), and Blue is at **33** (low strength). A lot of red, a fair bit of green, and a little blue blend together into a warm coral-orange — exactly the colour shown back on the # symbol page.

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Primary Colours & Mixing New Ones

When only one of the three lights is turned all the way up and the other two are off, we get a pure primary colour:



Full red, no green or blue



Full green, no red or blue

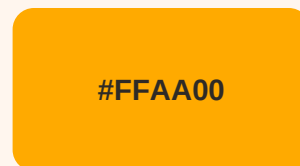


Full blue, no red or green

Mixing two lights together creates brand-new colours. For example, combining full red with a good amount of green — and leaving blue off — makes a warm orange:



Red, no green



Red + green = Orange

Pattern to remember: adding more of a second colour shifts the hue toward that colour; raising or lowering all three pairs together makes a colour lighter or darker.

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Designing a Digital Rainbow

By changing the hex values step by step, we can build an entire rainbow, one band at a time. Follow the sequence below from left to right:



Notice how Red (#FF0000) and Green (#00FF00) blend their way to Yellow (#FFFF00) before blue ever joins in — and Violet appears once red and blue mix at the very end. The whole rainbow is just the same three light dials, turned up and down in a smooth pattern.

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Activity Time: Guess the Colour!

Look at each hex code below. Before checking with a colour picker, write down what colour you predict it will be — then break down the R, G, and B pairs to check your guess.

#FF69B4 ?	My guess: _____
#1E90FF ?	My guess: _____
#00FF7F ?	My guess: _____

Bonus challenge: Without a colour picker, can you guess whether #444444 is a light grey or a dark grey? (Hint: compare it to #00 and #FF on the volume-dial scale!)

Hex colour codes are used everywhere in the digital world, not just in lessons like this one:

- **Website designers** use hex codes to choose exact colours for buttons, text, and backgrounds.
- **Game developers** use hex codes to colour characters, costumes, and entire game worlds.
- **App creators** rely on hex colours to make their apps look polished and consistent.
- **Digital artists** paint and design using hex codes to mix exactly the shade they imagine.

Next time you pick a colour in a drawing app or a slideshow tool, look for a small box that shows a code starting with #. Now you know how to read it!

What we learned today

- Hexadecimal counts using 16 digits: 0–9, then A–F.
- A # symbol tells a computer that a code describes a colour.
- A hex colour code has three pairs: Red, Green, and Blue.
- Each pair runs from 00 (none) to FF (full, equal to 255).
- Mixing R, G, and B in different amounts creates every colour on a screen.

Answer key

Hex digit practice (from page 2): 11 = B 14 = E 9 = 9 15 = F 12 = C

Guess-the-colour activity:

#FF69B4

Hot pink

#1E90FF

Dodger blue

#00FF7F

Spring green

#FF69B4 → R=FF(255) G=69(105) B=B4(180) | #1E90FF → R=1E(30) G=90(144) B=FF(255) | #00FF7F → R=00(0) G=FF(255) B=7F(127)

Bonus challenge: #444444 is a fairly dark grey — 44 in hex is 68 in decimal, which is much closer to 00 (none) than to FF (255), so very little light is glowing.

Great work exploring hex colours today! Keep experimenting with codes in any drawing app, website builder, or colour picker you can find — every code you try teaches your eyes to read the RGB pattern a little faster.